

Question Bank-----

Unit-1

- 1) Define Cloud Computing? Also Explain its characteristics, advantages and disadvantages?
- 2) Differentiate Distributed computing and cloud computing?
- 3) Discuss various cloud computing technologies on which cloud computing relies?
- 4) Discuss various cloud service and deployment models in detail.
- 5) Describe NIST Cloud reference architecture with diagram.
- 6) Explain and draw cloud computing architecture.
- 7) What is cloud migration and different cloud migration strategies?
- 8) Describe seven steps model for migrating into cloud.
- 9) Give a difference between traditional computing and cloud computing?
- 10) Give a difference between distributed computing and cloud computing?
- 11) Explain SOA(service oriented architecture).
- 12) Explain the evolution of cloud computing.
- 13) Explain Cloud and different types of cloud(deployment models)?
- 14) Explain cloud computing, Grid computing and distributed computing?

Unit-2

- 1) Explain virtualization and its characteristics.
- 2) Discuss various types of virtualization with example.
- 3) Elaborate various implementation levels of virtualization in detail.
- 4) Explain hypervisor and explain the working of type-1 and type-2.
- 5) Describe virtual machine migration and their types.
- 6) Explain all the steps of live migration with diagram.
- 7) Explain hypervisor? Draw and explain the working of type-1 and type-2 hypervisor.
- 8) Explain hardware virtualization and its types.
- 9) Define virtual cluster and its features.
- 10) A virtual machine 64 GB memory is migrated using pre-copy live migration. The available network bandwidth is 4 Gbps. The virtual machine has a dirty page rate of 400Mbps.(Perform at-least 3-4 migration iteration) Calculate-
 - a) Time for each migration iteration.
 - b) Amount of data transferred in each migration.
 - c) Downtime(assume final dirty memory=516MB)
- 11) Explain

(i) Virtualization structures (Xen Architecture, KVM)

(ii) Levels of virtualization.

11) Describe Memory Virtualization.

12) Describe I/O and CPU virtualization.

13) Analyze the impact of virtualization on data-center automation. Evaluate the advantages and challenges of using virtualization for managing large-scale cloud infrastructure, and propose an optimized strategy for resource management.

14) Explain resource management in cloud computing?